

# Time Series Forecasting of Product Demand

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## Abstract

This project investigates demand forecasting for Craft Beer, Whisky, and White Wine using classical time series forecasting methods. The work was inspired by the Skyrose Marketing Agency case study, which served solely as the project idea. The complete implementation, experimentation, model comparison, and analysis were performed independently to understand forecasting techniques and evaluate their practical performance.

## 1. Introduction

Demand forecasting helps organizations optimize inventory, marketing strategies, and operational planning. This project applies multiple statistical forecasting models to historical Google Trends data, compares their performance, and identifies the most suitable model for different demand patterns.

## 2. Dataset Description

- Source: Google Trends dataset used in the Skyrose case study.
- Products: Craft Beer, Whisky, and White Wine.
- Target Variable: Weekly Search Volume Index.
- Frequency: Weekly observations through 2025.
- Preprocessing: datetime conversion, removal of unnecessary columns, missing-value checks, and exploratory visualization.

## 3. Methodology

The following forecasting models were implemented and evaluated:

- Naive Forecast
- Moving Average / Weighted Moving Average
- Simple Exponential Smoothing
- Holt's Double Exponential Smoothing
- Holt-Winters Triple Exponential Smoothing
- ARIMA
- Auto-ARIMA

**Evaluation Metrics:** MSE, RMSE, and MAPE, followed by residual diagnostics.

## 4. Results & Evaluation

Naive Forecast provided the baseline. Moving Average reduced short-term fluctuations but had limited predictive capability. Holt's method effectively modeled trend, while Holt-Winters achieved the best performance for Craft Beer and White Wine by capturing both trend and seasonality. ARIMA performed particularly well for Whisky after stationarity adjustments, making Holt-Winters and ARIMA the strongest overall models.

## 5. Conclusion & Future Work

The project demonstrates that classical forecasting techniques remain effective for demand prediction when historical trend and seasonality are present. Future enhancements include SARIMA, Prophet, Random Forest, LSTM models, and incorporating external factors such as holidays and promotional events.

## Acknowledgement

**Project Inspiration:** Skyrose Marketing Agency Case Study.

This project was independently implemented for educational purposes. The Skyrose case study served only as the inspiration and problem statement; all coding, experimentation, analysis, and report preparation were completed independently.